



Question Set

Question 1

A Black-Scholes implied volatility

$$\sigma$$

solves

$$C_{BS}(\sigma) = C_{mkt}$$

Write Newton's method for sigma and explain why vega appears in the denominator.

Question 2

Do two Newton iterations to solve

$$x^2 - 2 = 0$$

from

$$x_0 = 1$$

Use this as an analogy for implied-volatility inversion.

Question 3

A least-squares calibration minimizes

$$F(\theta) = \frac{1}{2} \|r(\theta)\|^2$$

Derive the Gauss-Newton step.

Question 4

Explain why Levenberg-Marquardt damping is useful in volatility-surface calibration.

Question 5

For

$$f(x, y) = 100(y - x^2)^2 + (1 - x)^2$$

why is vanilla gradient descent difficult?

Question 6

Compute one gradient-descent step for $f(x, y) = (x-3)^2 + 10(y+1)^2$ from (0,0) with learning rate 0.05.

Question 7

Why can a calibration objective be small while the model is still bad for market risk?

Question 8

Derive the normal equations for minimizing

$$\|X\beta - y\|^2$$

Question 9

Explain line search in the context of Newton or quasi-Newton calibration.

Question 10

What is the difference between parameter bounds and parameter transformations? Give an example.

Question 11

A Hessian has eigenvalues 0.001 and 100. What does this imply for numerical optimization?

Question 12

Describe a robust calibration validation checklist after an optimizer converges.